

KUMLINGE KK, Finland

WMO: 027900

Lat: 60.258N

Lon: 20.747E

Elev: 74

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	12.1	0.9	5.8	7.4	6.0	7.5	13.4	32.1	34.2	28.4	32.4	8.8	80	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	75.5	66.6	72.7	64.5	70.1	63.2	68.7	73.3	66.5	71.2	64.5	68.7	10.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	99.2	71.6	64.5	91.2	69.4	62.6	85.0	67.1	32.9	73.4	31.1	71.0	29.6	68.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 28.3	24.9	22.2	DB	5.4	78.2	8.1	3.4	-0.4	80.6	-5.1	82.6	-9.7	84.5	-15.5	87.0	(4)
(5)			WB	4.2	68.8	7.6	3.0	-1.3	70.9	-5.7	72.7	-10.0	74.3	-15.6	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	29.2	27.6	31.1	39.1	48.9	56.3	63.8	63.0	55.9	46.2	39.5	34.7	(6)	
	DBStd	13.76	7.12	8.61	6.80	4.99	5.90	4.97	4.88	4.21	4.27	5.29	5.66	6.45	(7)	
	HDD50	3222	645	627	586	330	92	7	0	0	7	138	316	474	(8)	
	HDD65	7479	1110	1047	1051	778	498	262	84	87	273	584	766	939	(9)	
	CDD50	1297	0	0	0	2	59	197	429	405	185	20	0	0	(10)	
	CDD65	75	0	0	0	0	0	2	47	26	0	0	0	0	(11)	
	CDH74	160	0	0	0	0	1	5	122	32	0	0	0	0	(12)	
	CDH80	8	0	0	0	0	0	0	8	0	0	0	0	0	(13)	
(14) Wind	WSAvg	11.5	12.8	11.3	11.5	10.7	10.1	10.3	9.9	10.4	11.1	12.3	13.1	14.0	(14)	
(15) Precipitation	PrecAvg	32.9	3.4	2.5	2.1	1.7	1.7	2.4	2.9	3.1	2.8	3.3	3.4	3.7	(15)	
	PrecMax	40.9	5.8	5.4	3.9	2.8	3.6	4.4	5.4	5.7	5.6	6.7	7.8	6.5	(16)	
	PrecMin	24.3	0.5	0.1	0.4	0.2	0.4	1.0	0.2	1.1	0.8	0.5	0.5	0.7	(17)	
	PrecStd	4.0	1.5	1.3	0.8	0.7	0.8	0.9	1.3	1.2	1.5	1.5	1.5	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	41.9	47.4	59.1	70.2	73.6	80.8	77.8	67.5	58.4	50.3	46.9	(19)	
		MCWB	41.0	38.2	41.6	48.6	57.3	60.8	67.8	68.5	60.9	55.1	48.9	42.8	(20)	
	2%	DB	40.9	39.3	43.7	53.3	65.6	69.7	77.4	73.9	64.7	56.0	48.6	44.6	(21)	
		MCWB	39.5	37.1	39.2	45.3	55.6	59.4	67.5	65.8	59.5	53.5	47.4	41.6	(22)	
	5%	DB	39.5	38.0	40.8	49.3	61.5	66.8	74.3	71.4	63.0	54.3	47.5	43.1	(23)	
		MCWB	38.2	36.3	37.6	42.8	53.7	58.1	66.1	64.1	58.6	52.1	46.2	40.9	(24)	
	10%	DB	37.9	36.5	38.8	46.4	58.0	64.1	71.6	69.1	61.4	52.9	46.1	41.7	(25)	
		MCWB	36.5	35.2	36.4	41.1	51.3	56.9	64.9	62.7	57.8	50.8	44.8	40.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	39.1	42.2	49.4	60.1	63.9	71.6	71.4	62.5	56.4	49.7	45.0	(27)	
		MCDB	42.3	40.3	46.4	57.2	66.4	70.0	77.4	76.0	65.3	57.7	50.0	46.1	(28)	
	2%	WB	39.9	37.6	39.9	46.2	57.1	61.4	69.6	67.8	60.7	54.1	47.9	43.0	(29)	
		MCDB	40.6	38.7	42.5	52.4	63.8	67.0	74.2	72.9	63.7	55.4	48.4	43.8	(30)	
	5%	WB	38.5	36.4	38.1	43.8	54.3	59.7	67.8	65.6	59.5	52.7	46.5	41.5	(31)	
		MCDB	39.3	37.6	40.1	48.3	60.6	64.9	72.3	69.4	62.0	54.0	47.3	42.5	(32)	
	10%	WB	36.6	35.2	36.6	41.6	51.9	57.8	65.9	64.0	58.5	51.3	45.0	40.2	(33)	
		MCDB	37.7	36.3	38.5	45.4	57.5	62.6	71.0	67.6	60.9	52.7	45.9	41.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	8.1	9.6	10.5	9.5	8.5	7.8	6.5	5.4	4.7	5.2	(35)	
		MCDBR	5.6	6.0	9.6	14.7	16.8	14.4	12.4	10.6	8.0	6.0	4.7	5.5	(36)	
	5% WB	MCWBR	5.4	5.4	6.7	8.8	9.0	6.8	5.8	5.1	4.7	4.9	4.8	5.5	(37)	
		MCWBR	5.5	5.6	9.1	13.9	15.8	12.6	10.8	8.9	7.3	5.8	4.6	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.275	0.310	0.329	0.368	0.366	0.357	0.386	0.376	0.347	0.332	0.300	0.386	(40)	
		taud	2.177	2.251	2.304	2.329	2.424	2.492	2.437	2.452	2.516	2.413	2.219	2.649	(41)	
	Ebn at Noon	Ebn at Noon	154	216	252	263	273	278	265	259	248	209	141	108	(42)	
		Edh at Noon	13	22	28	33	32	31	32	29	23	19	13	9	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	74	265	758	1282	1759	1939	1801	1402	856	363	99	46	(44)	
	RadStd	RadStd	10	47	85	122	150	128	160	121	97	54	20	6	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	-198	N/A	N/A	N/A	N/A

Nomenclature: See separate page